

Questions

1. What does **image classification** mean? Explain with one example.
2. Why is using a **Dense Network** not always a suitable choice for images?
3. In a CNN, what is **convolution**, and what does a filter or kernel do?
4. What is a **feature map** in a convolutional neural network?
5. Why is **weight sharing** used in CNNs? Explain.
6. How does the **ReLU activation function** work? Write the ReLU output for the following values:
[-3, 0, 2, 5]
7. What is **max pooling**, and why is it used in CNNs?
8. In the Fashion-MNIST code, why are images reshaped from 28×28 to $28 \times 28 \times 1$?
9. What is the difference between **training data, validation data, and test data**?
10. What is **overfitting**? If training accuracy is high but validation accuracy is low, what does this indicate?